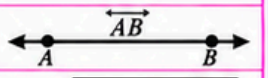
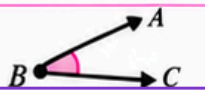
Geometry Basics Cheat Sheet

Unit Focus: points, lines, rays, segments, and angles

1 KEY VOCABULARY

point	An exact location in space.	
line	A straight path that extends forever in both directions.	
plane	A flat surface that extends forever in all directions.	
segment	A part of a line with two endpoints.	
ray	A part of a line with one endpoint that extends forever in one direction.	
angle	A figure formed by two rays with a common endpoint (vertex).	

collinear: points on the same straight line

coplanar: points on the same plane

2 NAMING AND DRAWING FIGURES

Use capital letters to name geometric figures.

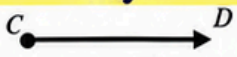
Line Segment



Segment $\overline{AB}$

$\overline{AB}$

Ray



Ray $\overrightarrow{CD}$

$\overrightarrow{CD}$

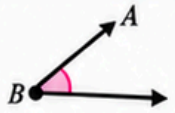
Line



Line EF

$\overleftrightarrow{EF}$

Angle



Angle ABC (vertex is B)

$\angle ABC$

IMPORTANT: Ray order matters.

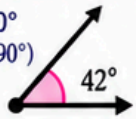
Ray CD differs from ray DC; angle ABC = CBA.

3 CLASSIFYING ANGLES

Angles are classified by their measures.

Acute Angle

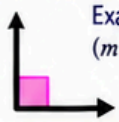
Less than 90° ($0° < m\angle < 90°$)



Example: 42° is an acute angle.

Right Angle

Exactly 90° ($m\angle = 90°$)



Obtuse Angle

Greater than 90° but less than 180° ($90° < m\angle < 180°$)



Straight Angle

Exactly 180° ($m\angle = 180°$)

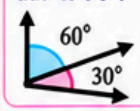


4 ANGLE RELATIONSHIPS

Special relationships between pairs of angles.

Complementary Angles

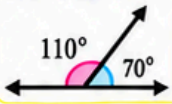
Two angles whose measures add to 90°.



$m\angle 1 + m\angle 2 = 90°$ (are complementary)

Supplementary Angles

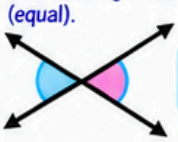
Two angles whose measures add to 180°.



$m\angle 1 + m\angle 2 = 180°$ (are supplementary)

Vertical Angles

Opposite angles formed by intersecting lines are congruent (*equal*).



Vertical angles are congruent.

Linear Pair

Two adjacent angles that form a straight line are supplementary (their measures add to 180°).

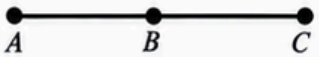


Example: If one angle in a linear pair is 68°, the other is 112°.

5 SEGMENT ADDITION AND MIDPOINT

Segment Addition Postulate

If point B is between A and C, then $AB + BC = AC$.



$AB + BC = AC$

Midpoint

A midpoint divides a segment into two congruent segments (equal lengths).



Example: If $AB = 18$ and M is the midpoint of AB, then $AM = MB = 9$.

6 USING ALGEBRA WITH ANGLES

Example: Solving for an angle measure

The angles $(x + 15)°$ and $(2x - 3)°$ are supplementary.

Step 1: Set up the equation.
$(x + 15) + (2x - 3) = 180$

Step 2: Combine like terms.
$3x + 12 = 180$

Step 3: Solve for x.
$3x = 168$
$x = 56$

Find the angle measures.

$x = 56$

First angle: $x + 15 = 56 + 15 = 71°$

Second angle: $2x - 3 = 2(56) - 3 = 112 - 3 = 109°$

Answer: The angles are 71° and 109°.

7 CONGRUENT FIGURES

Congruent segments have equal lengths.

Congruent angles have equal measures.

Use matching tick marks and angle arcs.

8 QUICK REMINDERS & COMMON MISTAKES

DO REMEMBER:

- Use degree symbols (°) with angle measures.
- Vertical angles are congruent (equal).
- A midpoint creates two equal segments.
- Decide whether angles should add to 90° or 180° (complementary or supplementary).
- Use correct notation and order when naming rays and angles.

COMMON MISTAKES:

- Forgetting to use degree symbols.
- Thinking vertical angles are supplementary (they are congruent).
- Not dividing a segment into equal parts at the midpoint.
- Adding angles to the wrong total (90° instead of 180°, or vice versa).
- Writing ray or angle names in the wrong order.

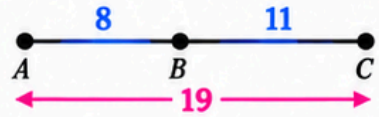
Geometry Basics Cheat Sheet

More Practice: *measuring, naming, and angle relationships*

1 SEGMENTS & MIDPOINTS

Segment Addition Postulate

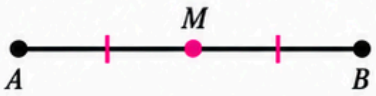
If point B is between A and C, then $AB + BC = AC$.



Example:
If $AB = 8$ and $BC = 11$, then $AC = 8 + 11 = 19$.

Midpoint

A midpoint divides a segment into two congruent segments (equal lengths).



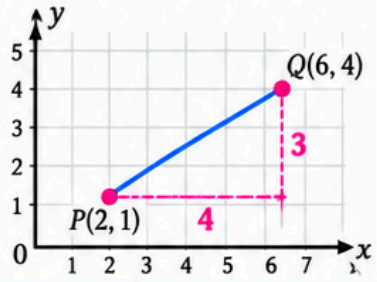
Example:
If $AB = 18$ and M is the midpoint, then $AM = MB = 9$.

2 DISTANCE & COORDINATES

The distance between two points (x_1, y_1) and (x_2, y_2) is

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Example: Find the distance between $P(2, 1)$ and $Q(6, 4)$.

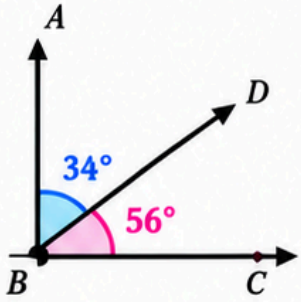


Use the distance formula.

$$d = \sqrt{(6-2)^2 + (4-1)^2}$$
$$= \sqrt{16 + 9}$$
$$= \sqrt{25}$$
$$= 5 \text{ units}$$

3 ANGLE ADDITION

If ray BD is inside $\angle ABC$, then $m\angle ABD + m\angle DBC = m\angle ABC$.



Example:

$m\angle ABD = 34°$
$m\angle DBC = 56°$
$m\angle ABC = 90°$

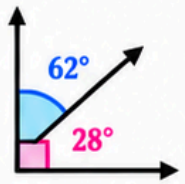
$34° + 56° = 90°$
$m\angle ABC = 90°$

4 COMPLEMENTARY & SUPPLEMENTARY

Two special relationships between pairs of angles.

Complementary Angles

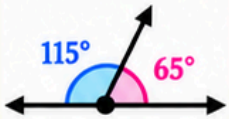
Two angles whose measures add to 90°.



$28° + 62° = 90°$

Supplementary Angles

Two angles whose measures add to 180°.



$115° + 65° = 180°$

Other Relationships

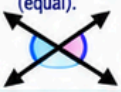
Linear Pair

Two adjacent angles that are supplementary (their measures add to 180°).



Vertical Angles

Opposite angles formed by intersecting lines are congruent (equal).



Vertical angles are congruent.

5 POINT, LINE, PLANE

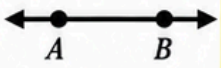
Point

An exact location in space.

• A

Line

A straight path that extends forever in both directions.



Plane

A flat surface that extends forever in all directions.



- **Collinear points:** Points that lie on the same line.
- **Coplanar points:** Points that lie in the same plane.
- **Intersecting lines:** Lines that cross at exactly one point.
- **Perpendicular lines:** Lines that intersect to form a 90° angle.

6 NAMING FIGURES

Line Segment



Segment $\overline{AB}$ can also be written as segment BA.
$\overline{AB} = BA$

Ray

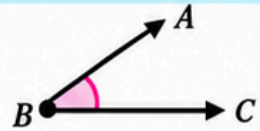


Ray AB starts at A and passes through B.

Ray BA starts at B and passes through A.

Order matters for rays.

Angle



An angle can be named using three letters with the vertex in the middle.

$\angle ABC$ and $\angle CBA$ name the same angle because they refer to the same rays and B is the vertex.

7 QUICK REMINDERS & COMMON MISTAKES

DO REMEMBER:

- Name angles with the vertex in the middle (e.g., $\angle ABC$).
- Midpoints split segments into two equal halves.
- Sketch and label points in order from left to right (or as given).
- Use degree symbols (°) with angle measures.
- Check your work and make clear, labeled diagrams.

COMMON MISTAKES:

- Reversing the order of endpoints for a ray (ray AB is not the same as ray BA).
- Confusing congruent angles with supplementary angles (congruent means equal measure, not sum to 180°).
- Forgetting to put the vertex in the middle when naming angles.
- Leaving out degree symbols (e.g., writing 90 instead of 90°).
- Not labeling diagrams clearly.